1 **(a)** $\dfrac{V}{t} = \dfrac{\pi P r^4}{8 C l}$

$C = [\pi \times 2.5 \times 10^3 \times (0.75 \times 10^{-3})^4] / (8 \times 1.2 \times 10^{-6} \times 0.25)$ C1

$= 1.04 \times 10^{-3}\ \mathrm{N\,s\,m^{-2}}$ A1 [2]

(b) $4 \times \%r$ C1

$\%C = \%P + 4 \times \%r + \%V/t + \%l$

$= 2\% + 5.3\% + 0.83\% + 0.4\%\ (= 8.6\%)$ A1

$\Delta C = \pm\ 0.089 \times 10^{-3}\ \mathrm{N\,s\,m^{-2}}$ A1 [3]

(c) $C = (1.04 \pm 0.09) \times 10^{-3}\ \mathrm{N\,s\,m^{-2}}$ A1 [1]

2 **(a)** **(i)** $v^2 = u^2 + 2as$